Auditable Structure-Aware Transfer of QAOA Parameters Across Biomedical Weighted-Graph Families

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This paper studies whether fixed-depth QAOA parameters can be transferred across related weighted instances by learning a map from graph structure to variational angles, and whether such transfer survives a genuinely independent family shift. The primary test bed is a transcriptomic weighted-MaxCut benchmark built from a cached prostate-expression cohort, where each instance is a sparse co-expression graph and each reference target is generated by exact depth-2 statevector evaluation combined with multistart classical search. On a 16-gene held-out transcriptomic family, the graph-conditioned initializer attains mean approximation ratio 0.8623 versus 0.8628 for direct search while reducing median online latency from 19687.1 ms to 12.17 ms, corresponding to 99.94% retention of the direct-search objective. The paper then adds a second independent weighted graph family derived from breast-cancer morphology descriptors. On this 12-feature morphology family, within-family morphology initializers remain highly concentrated: the morphology heuristic and morphology-family oracle GNN attain 99.76% and 99.35% retention of the corresponding classical ceiling, respectively. In contrast, a transcriptomic-trained GNN transferred without retraining to the morphology family collapses to 67.23% retention, even though the transcriptomic family heuristic still attains 98.90% retention. The paper connects that failure to a pre-objective angle-excursion diagnostic: the transferred predictions lie 2.03 radians from the source-family centroid, implying a 1.75-radian lower bound on target angle error before evaluating the quantum objective, and the observed mean target-angle error is 2.14 radians. Complementary reruns show transcriptomic seed robustness on the reduced 10-gene benchmark, nonmonotone sensitivity to adaptation-family size, a bounded real-data size window, smooth degradation under local depolarizing noise, and additional transcriptomic family-shift failures at 10 and 14 genes. These observations are supported by a second-order regret bound, an expressivity separation from affine graph-feature regression, a fixed-family non-plateau statement, an angle-excursion proposition for transfer failure, a depolarizing-noise contraction theorem, and a corollary showing exact contraction of pairwise objective gaps under local depolarizing noise. The resulting picture is sharper and more limited: within-family amortized search is strong on concentrated biomedical graph families, whereas out-of-family failure can be anticipated once transferred predictions exit the source concentration region.

I. INTRODUCTION

The Quantum Approximate Optimization Algorithm (QAOA) remains one of the central fixed-depth paradigms for near-term combinatorial optimization because shallow variational circuits can retain interpretable structure tied to the underlying cost Hamiltonian [1, 2]. That same structure does not make parameter selection easy: even low-depth instances still require careful outer-loop optimization, and the extent to which good parameters transfer across related graphs remains strongly family dependent [3, 4].

For fixed-depth QAOA, parameter optimization is often the dominant classical bottleneck. Even when only a small number of angles must be selected, each new graph instance ordinarily requires a fresh outer-loop search, and repeated deployment on a related weighted family can spend far more time searching for parameters than evaluating the resulting circuit [1, 3]. This cost becomes especially pronounced when one repeatedly solves weighted graphs drawn from a common structured family, as occurs in transcriptomic and other data-derived optimization settings [2, 4].

The key gap is therefore not the existence of QAOA heuristics, but the lack of a well-audited bridge between instance-family structure and instance-dependent parameter prediction. Graph learning provides a natural language for such a bridge because message passing and graph-level readout can encode weighted relational structure while preserving the permutation symmetries of the instance family [5, 6]. At the same time, graph-benchmark practice has shown that structural claims need disciplined baselines and tightly controlled evaluation rather than architecture-heavy optimism [7, 8]. A convincing contribution in this setting therefore requires a clearly specified graph family, a precise downstream quantum objective, and evidence showing where learned transfer remains stable and where it fails.

This paper addresses that gap by learning QAOA parameters from graph structure. Fixed-depth parameter selection is modeled as a graph-to-angle map and a compact graph neural network predicts the depth-2 QAOA angles $(\gamma_1, \gamma_2, \beta_1, \beta_2)$ for weighted MaxCut instances derived first from transcriptomic co-expression graphs and then from an independent morphology-correlation family built from breast-cancer descriptors. The model choice is intentionally conservative: recent graph-learning evidence suggests that strong baselines often remain simple when the protocol is tightly controlled [9, 10]. The resulting angles are evaluated only through the downstream quantum objective, allowing the learned predictor to be judged directly against multi-start classical search

on held-out graphs. The manuscript now uses a single extracted-script evidence path for the headline transcriptomic result and then probes that same code path with stronger baselines, transcriptomic family shift, independent cross-domain transfer, seed variation, size variation, adaptation-family-size variation, and noise. Taken together, these experiments support a narrower but better substantiated claim: graph-conditioned prediction can amortize fixed-depth QAOA search on concentrated weighted graph families, but its failure can be abrupt once transferred predictions leave the source-family concentration region.

The paper makes four concrete contributions:

- The paper formulates fixed-depth QAOA parameter transfer as a structure-aware map $\theta^\star(G) \approx f_\phi(G)$ from weighted graphs to variational angles and relates that map to the underlying cost Hamiltonian.
- The paper presents two auditable biomedical weighted-MaxCut families, one transcriptomic and one morphology-derived, in which every held-out claim is checked against exact MaxCut values and exact depth-2 statevector evaluation.
- The paper provides theorem-level support through a second-order regret bound, an expressivity separation from affine graph-feature regression, a finite-family trainability statement, and an angle-excursion diagnostic that lower-bounds target angle error from pre-objective prediction geometry.
- The paper extends the empirical evidence with an extracted-script headline benchmark, a stronger within-family baseline suite, a transcriptomic family-shift study, and a stronger independent cross-domain transfer study in addition to seed robustness, real-data size scaling, and adaptation-family-size sensitivity.

II. BACKGROUND AND PRIOR WORK

The relevant prior literature has four ingredients. First, fixed-depth QAOA studies show both positive evidence for transferable parameter regularities and clear limits on when those regularities survive family shift [1, 2]. Complementary analyses of transfer and counterdiabatic variants make clear that useful structure exists, but that it is sensitive to graph family, depth, and optimization protocol [3, 4].

Second, the graph-learning side of the problem is now mature enough that graph-level encoders can be chosen from a well-understood design space rather than introduced as bespoke black boxes. Neighborhood aggregation and message passing are well established [5, 11], and subsequent work has clarified tradeoffs involving expressive graph representations, deeper propagation, and architecture scaling [12, 13]. The present manuscript uses that literature in a deliberately conservative way: it asks whether a compact graph encoder is sufficient for the audit-oriented transfer problem posed here, not whether another layer of architectural complexity can improve one benchmark [9, 14].

Third, benchmark discipline matters as much as model class. Public graph-learning evidence has repeatedly shown that apparent model gains can shrink once protocols, baselines, and dataset construction are standardized [7, 15]. Design-space studies and graph-specific benchmark analyses reinforce the same lesson: strong conclusions require comparison against simple competitive baselines, not only against weak random or undertuned alternatives [8, 16].

Fourth, the learning component is closest to predict-then-optimize and learned optimization rather than to end-to-end classification. In this setting, a predictor is useful only insofar as it preserves the downstream objective induced by the angles it outputs [17, 18]. That viewpoint aligns naturally with differentiable and decision-focused optimization formulations in which surrogate prediction quality is secondary to the final objective value [19, 20]. It also motivates the stronger baseline suite used here, because recent work in learned optimization and structured benchmarks has shown that simple non-neural methods can remain highly competitive on finite-family problems [21, 22].

Within graph-level model design, the relevant comparison class remains broad. Pooling architectures, attention-based graph networks, transformer-style graph encoders, and contrastive pretraining all expand the available hypothesis class [23, 24]. Yet current evidence also indicates that simpler graph-convolution baselines and carefully benchmarked architectures can be surprisingly hard to beat on medium-scale graph tasks [25, 26]. Related work on graph explanation and graph-level prediction likewise shows that evaluation artifacts can masquerade as structural understanding when the protocol is weak [27, 28]. This makes the present near-parity and failure-regime findings scientifically useful rather than incidental: when a compact encoder ties or only marginally exceeds simpler within-family rules, that is information about the instance family itself [29, 30].

III. THEORETICAL MOTIVATION AND PROBLEM FORMULATION

Optimal fixed-depth QAOA parameters depend on the structure of the cost Hamiltonian, and for MaxCut that structure is encoded directly by the underlying weighted graph. Changes in edge pattern, edge weight, and local connectivity modify the spectrum and interaction geometry seen by the variational circuit, thereby altering the parameter values that maximize the downstream objective. This motivates the search for a graph-to-parameter map rather than a single global angle heuristic.

Graph neural networks are a natural model class for

this setting because they process graph topology, node-level information, and weighted neighborhood interactions through permutation-equivariant message passing. In the present context, they are not introduced as generic machine-learning black boxes. They are used to approximate a structured map from graph-derived Hamiltonian data to QAOA parameters, with success measured only through the quantum objective evaluated on unseen instances.

We consider weighted MaxCut on a graph $G = (V, E, w)$ with $|V| = n$. The corresponding cost Hamiltonian is

$$H_C(G) = \frac{1}{2} \sum_{(i,j)\in E} w_{ij}(I - Z_i Z_j), \tag{1}$$

and the standard mixer Hamiltonian is

$$H_M = \sum_{i=1}^{n} X_i. \tag{2}$$

At depth p, the QAOA ansatz is

$$|\psi_p(\boldsymbol{\gamma}, \boldsymbol{\beta}; G)\rangle = \prod_{k=1}^{p} e^{-i\beta_k H_M} e^{-i\gamma_k H_C(G)} |+\rangle^{\otimes n}. \tag{3}$$

For clarity we write the layer unitaries explicitly as

$$U_C(\gamma) = e^{-i\gamma H_C}, \qquad U_M(\beta) = e^{-i\beta H_M}. \tag{4}$$

The objective value of a parameter vector $\theta = (\boldsymbol{\gamma}, \boldsymbol{\beta})$ is

$$C_G(\theta) = \langle \psi_p(\theta; G) | H_C(G) | \psi_p(\theta; G)\rangle, \tag{5}$$

and its approximation ratio is

$$r_G(\theta) = \frac{C_G(\theta)}{C^*(G)}, \tag{6}$$

where $C^*(G)$ denotes the exact MaxCut value.

The learned initializer is a graph-conditioned map

$$f_\phi : \mathcal{G} \to \mathbb{R}^{2p}, \qquad f_\phi(G) = \hat{\theta}_G. \tag{7}$$

The underlying scientific hypothesis can therefore be written as

$$\theta^\star_G \approx f_\phi(G), \tag{8}$$

where $\theta^\star_G$ denotes a high-quality fixed-depth optimizer for graph G. The role of learning is not to replace the Hamiltonian formulation but to approximate this instance-dependent map from weighted graph structure to useful QAOA parameters.

In the QAOA branch, $p = 2$ and

$$\hat{\theta}_G = (\hat{\gamma}_1, \hat{\gamma}_2, \hat{\beta}_1, \hat{\beta}_2). \tag{9}$$

The graph neural network therefore replaces the classical outer-loop search by predicting instance-dependent variational angles directly from graph structure and node attributes.

In the training setup, the network is fit against reference parameters $\theta^\star_G$ obtained by multi-start classical optimization on the adaptation graphs. The empirical training objective is the angle-regression loss

$$\mathcal{L}_{\text{train}}(\phi) = \frac{1}{|\mathcal{D}_{\text{adapt}}|} \sum_{G\in\mathcal{D}_{\text{adapt}}} \|f_\phi(G) - \theta^\star_G\|_2^2. \tag{10}$$

Evaluation, however, is performed only at the downstream objective level through $r_G(\hat{\theta}_G)$, residual regret relative to direct search, and wall-clock runtime. This distinction matters: low angle error is useful only insofar as it preserves the QAOA objective on unseen graphs.

A. Formal Results

The following results formalize the central mechanism studied in the experiments: graph structure determines the optimization landscape, prediction error induces downstream regret only at second order near a local optimum, and the fixed-depth regime considered here admits explicit trainability and noise-robustness statements.

Theorem 1 (Second-order regret bound)**.** *Let $\theta^\star_G$ be a local maximizer of $C_G(\theta)$ for a fixed graph G, and suppose the Hessian of C_G satisfies*

$$\|\nabla^2 C_G(\theta)\|_2 \le L_G \tag{11}$$

for all θ on the line segment between $\theta^\star_G$ and a prediction $\hat{\theta}_G$. Then

$$C_G(\hat{\theta}_G) \ge C_G(\theta^\star_G) - \frac{L_G}{2}\|\hat{\theta}_G - \theta^\star_G\|_2^2, \tag{12}$$

and therefore

$$r_G(\hat{\theta}_G) \ge r_G(\theta^\star_G) - \frac{L_G}{2C^*(G)}\|\hat{\theta}_G - \theta^\star_G\|_2^2. \tag{13}$$

Proof. Taylor expansion around $\theta^\star_G$ gives

$$C_G(\hat{\theta}_G) = C_G(\theta^\star_G) + \nabla C_G(\theta^\star_G)^\top \Delta + \frac{1}{2}\Delta^\top \nabla^2 C_G(\xi_G)\Delta, \tag{14}$$

where $\Delta = \hat{\theta}_G - \theta^\star_G$ and ξ_G lies on the segment joining the two points. Because $\theta^\star_G$ is stationary, the first-order term vanishes. Using the operator-norm bound yields

$$C_G(\hat{\theta}_G) \ge C_G(\theta^\star_G) - \frac{L_G}{2}\|\Delta\|_2^2. \tag{15}$$

Dividing by $C^*(G)$ gives the ratio bound. □

Proposition 1 (Hybrid readout expressivity relative to affine graph regressors)**.** *Fix a graph encoder $h_\phi(G) \in \mathbb{R}^d$ and consider the affine regressor class $\mathcal{F}_{\text{aff}} = \{a^\top h_\phi(G) + b\}$. Let the hybrid readout class be obtained by predicting QAOA angles through an affine map $\theta(G) =$*

$Wh_\phi(G)+b$ and evaluating the exact depth-1 single-edge MaxCut objective. Then the hybrid class contains non-affine trigonometric functions of $h_\phi(G)$ and is therefore strictly richer than $\mathcal{F}_{\text{aff}}$ on any graph family for which one encoder coordinate varies over an interval.

Proof. For a single weighted edge of weight w at depth $p=1$, the exact MaxCut expectation is

$$C(\gamma,\beta)=\frac{w}{2}\left(1+\sin(4\beta)\sin(2\gamma)\right). \tag{16}$$

Fix $\beta=\pi/8$ and let one coordinate of $Wh_\phi(G)+b$ equal $\gamma=u(G)$. Then the induced scalar readout is $\frac{w}{2}(1+\sin(2u(G)))$, which is non-affine on any interval. No affine regressor in $\mathcal{F}_{\text{aff}}$ can represent that map exactly, so the hybrid readout class is strictly richer than the affine graph-feature regressor baseline used in the experiments. □

Proposition 2 (Finite-family non-plateau statement)**.** *Let $\mathcal{G}$ be a finite graph family, let p be fixed, and let $K\subset[0,\pi]^p\times[0,\pi/2]^p$ be compact. If K excludes an ε-neighborhood of the common stationary set*

$$\mathcal{S}=\bigcap_{G\in\mathcal{G}}\{\theta:\nabla C_G(\theta)=0\}, \tag{17}$$

then there exists $c_{\varepsilon,K}>0$ such that

$$\min_{\theta\in K\setminus N_\varepsilon(\mathcal{S})}\frac{1}{|\mathcal{G}|}\sum_{G\in\mathcal{G}}\|\nabla C_G(\theta)\|_2^2\geq c_{\varepsilon,K}. \tag{18}$$

Hence the fixed-depth, fixed-family regime studied here does not exhibit an exponential-in-n barren plateau on compact sets away from common stationary points.

Proof. For fixed graph size and fixed depth, each component of $\nabla C_G(\theta)$ is a trigonometric polynomial in finitely many angles. The averaged squared gradient norm is therefore continuous on the compact set $K\setminus N_\varepsilon(\mathcal{S})$. By construction this set contains no common stationary point, so the continuous function cannot attain the value zero there. Compactness then implies a positive minimum $c_{\varepsilon,K}$. □

Proposition 3 (Angle-excursion diagnostic for transfer failure)**.** *Let the target-family optimizers lie in a ball $B(\mu_t,r_t)$ in angle space, and let μ_s be the source-family centroid. For any transferred prediction $\hat\theta$ on a target instance with optimizer $\theta_t^\star\in B(\mu_t,r_t)$,*

$$\|\hat\theta-\theta_t^\star\|_2\geq\|\hat\theta-\mu_s\|_2-\|\mu_s-\mu_t\|_2-r_t. \tag{19}$$

Consequently, if

$$\|\hat\theta-\mu_s\|_2>\|\mu_s-\mu_t\|_2+r_t+\delta, \tag{20}$$

then the target-angle error satisfies $\|\hat\theta-\theta_t^\star\|_2>\delta$ before the quantum objective is evaluated.

Proof. By the triangle inequality,

$$\|\hat\theta-\theta_t^\star\|_2\geq\|\hat\theta-\mu_s\|_2-\|\mu_s-\mu_t\|_2-\|\mu_t-\theta_t^\star\|_2. \tag{21}$$

Using $\|\mu_t-\theta_t^\star\|_2\leq r_t$ yields the first claim, and the second follows immediately. □

Theorem 2 (Depolarizing-noise contraction for MaxCut expectation)**.** *Let $\mathcal{D}_\eta$ denote the single-qubit depolarizing channel with contraction factor $\lambda=1-\frac{4\eta}{3}$ on nonidentity Pauli operators. Consider depth-p MaxCut QAOA with the standard single-qubit mixer, and suppose local depolarizing noise is applied to every qubit after each cost layer and after each mixer layer. Then for every weighted graph G and parameter vector θ,*

$$C_{G,\eta}(\theta)-\frac{W(G)}{2}=\lambda^{4p}\left(C_{G,0}(\theta)-\frac{W(G)}{2}\right), \tag{22}$$

where $W(G)=\sum_{(i,j)\in E}w_{ij}$.

Proof. Write

$$H_C(G)=\frac{W(G)}{2}I-\frac{1}{2}\sum_{(i,j)\in E}w_{ij}Z_iZ_j. \tag{23}$$

The identity term is invariant. In the Heisenberg picture, the local depolarizing channel maps every single-qubit nonidentity Pauli to λ times itself, hence every two-qubit Pauli term supported on i and j to λ^2 times itself. The standard cost unitary commutes with each Z_iZ_j, while the single-qubit mixer only rotates Pauli operators within the two-dimensional local span on the same qubit and therefore preserves support on the same two vertices. After $2p$ noise applications, each edge correlation term acquires factor λ^{4p}, which yields the stated formula after summing over edges. □

Corollary 1 (Exact contraction of pairwise objective gaps under local depolarizing noise)**.** *Under the hypotheses of the depolarizing-noise contraction theorem, for any two parameter vectors θ and θ', one has*

$$C_{G,\eta}(\theta)-C_{G,\eta}(\theta')=\lambda^{4p}\left(C_{G,0}(\theta)-C_{G,0}(\theta')\right). \tag{24}$$

Hence every pairwise objective gap contracts by the same multiplicative factor λ^{4p}.

Proof. Apply the depolarizing-noise contraction theorem separately to θ and θ' and subtract the two identities. The constant offset $W(G)/2$ cancels exactly. □

Remark 1. *The second-order regret bound, angle-excursion proposition, depolarizing-noise contraction theorem, and pairwise-gap corollary specify how parameter error, pre-objective geometric mismatch, and local noise propagate in the fixed-depth learned-initialization regime studied here.*

IV. METHODS

This section states the weighted-graph benchmarks and learning pipeline at the level needed for direct reproduction from the repository. A central point for the present revision is that the main claim now rests on extracted-script artifacts rather than on notebook-only outputs. The same exact-audited code path is reused for stronger within-family baselines, transcriptomic family-shift studies, an independent morphology-family transfer study, seed and adaptation sensitivity checks, and the noisy-QAOA and size-sweep reruns. This emphasis on one auditable protocol follows the lesson from graph benchmarking that evaluation drift can otherwise dominate model-level conclusions [7, 15].

A. Transcriptomic panel and graph construction

The optimization benchmark uses transcriptomic co-expression graphs derived from a cached prostate-expression cohort. The code preferentially loads a cached top-32 variance-ranked panel and falls back to a legacy top-10 cache if needed; the extracted headline benchmark uses a 16-gene variance-ranked panel, while the reduced reruns use smaller panels when exact auditing would otherwise become too expensive. Given a selected gene panel, the graph builder computes absolute Pearson correlations on a patient subsample, forms a maximum spanning tree, and then adds the strongest remaining correlations until a target edge budget is reached. For the extracted 16-gene headline benchmark this procedure yields connected 18-edge weighted graphs; for the reduced cross-family transfer study, the 10-gene target family uses 18 edges and the 14-gene target family uses 36 edges.

Each graph instance is therefore a sparse weighted MaxCut problem whose vertices correspond to transcriptomic features and whose edge weights encode correlation strength. The held-out instances are not arbitrary graph draws. They are patient-resampled views of the same cached expression artifact, produced by split-safe stratified subsampling under fixed seeds. This is exactly the regime in which parameter transfer is scientifically meaningful: the instances are related but not identical, and exact downstream evaluation remains feasible. The setup is intentionally closer to a controlled graph-family benchmark than to a broad biological generalization claim [8, 15].

For every graph instance the code computes the full cut diagonal and the exact MaxCut value by brute force. This exactness is important. It means that approximation-ratio claims are not inferred indirectly from sampling surrogates or approximate combinatorial solvers, but are checked directly against the exact objective on every held-out graph.

To move beyond transcriptomic panel construction, we added a second weighted family derived from the checked-in breast-cancer morphology table. Because the table contains standardized continuous descriptors rather than class labels, we rank features by mean absolute correlation over the full cohort, keep the top 12 descriptors, and then rebuild a sparse 18-edge correlation graph from each 100-patient uniform subsample. The resulting morphology instances are built by exactly the same maximum-spanning-tree-plus-strongest-edges rule used for the transcriptomic family, so the family shift is in the underlying data source rather than in the graph-construction mechanism. This gives a clean second biomedical family in which exact depth-2 auditing remains feasible while the domain is clearly outside the original transcriptomic panel.

B. Reference solver, learned predictor, and baselines

The reference optimizer is a multistart Nelder–Mead search over the depth-2 angle vector. For each instance, the code launches several restarts from random initial angles, normalizes the resulting parameters into the canonical QAOA ranges $\gamma_k \in [0, \pi)$ and $\beta_k \in [0, \pi/2)$, and keeps the best solution by exact statevector value. In the default benchmark configuration this search uses eight restarts and 320 iterations per restart, matching the fixed-depth exact-simulation regime in which parameter-transfer claims can be audited most cleanly [1, 3].

The learned predictor is the repository's SimpleGCN architecture. It performs two rounds of graph propagation, averages node embeddings by global mean pooling, and maps the pooled representation to four real outputs corresponding to $(\gamma_1, \gamma_2, \beta_1, \beta_2)$. This puts the model squarely in the neighborhood-aggregation tradition of graph representation learning rather than in a custom problem-specific architecture family [5, 9]. The node feature used in the extracted transcriptomic code is intentionally simple: a single scalar equal to the row sum of the self-looped weighted adjacency. The point of the benchmark is therefore not to maximize architectural sophistication, but to test whether even a compact graph-conditioned predictor can replace repeated fixed-depth search on a structured weighted family [8, 10].

Training uses the angle-regression objective introduced above, with Adam optimization, early stopping on training loss, and exact classical references as targets. Evaluation is always downstream. The learned angles are inserted into the exact depth-2 QAOA circuit and judged by expected cut value, approximation ratio, and runtime. This is a predict-then-optimize protocol: the point is not to minimize surrogate regression error in isolation, but to retain downstream objective value after the learned angles are deployed [20, 31]. The primary nonlearned baseline is direct classical search, because that is the operational ceiling any inference-time replacement must match. The extracted headline benchmark adds a stronger baseline suite: a heuristic mean-angle

initializer, a descriptor k-nearest-neighbor regressor, an affine graph-feature regressor, and inference-time ablations that remove graph edges or node features. Including these simpler baselines is important because strong non-neural competitors often remain difficult to beat on structured finite-family problems [32, 33]. More broadly, the modern tabular-baseline literature cautions against assuming that a neural model should dominate simpler regressors by default [34, 35], and even attentive tabular architectures need careful comparison rather than presumption of superiority [33, 36].

Figure 1 summarizes this pipeline. The scientific claim is not that a graph neural network predicts angles accurately in a regression sense. The claim is that graph-conditioned prediction can retain the downstream QAOA objective that direct search would otherwise provide.

C. Extracted headline benchmark and auxiliary reruns

The paper uses a single extracted benchmark pipeline for the headline transcriptomic result. This pipeline reconstructs the 16-gene transcriptomic family, fits the graph model under fixed hyperparameters, and evaluates the stronger baseline suite used in the main text. Companion reruns then probe seed sensitivity, adaptation-family-size sensitivity, noisy evaluation, a reduced real-data size sweep, a reduced transcriptomic family-shift study, and an independent morphology-transfer study. The scientific advantage of this organization is that every main-text claim comes from the same auditable execution path.

V. EXTENDED EXPERIMENTAL DESIGN

The extracted headline benchmark and the auxiliary rerun studies are designed to answer slightly different questions. The headline benchmark asks whether graph-conditioned prediction can match direct search on a biologically derived held-out family under a fully scripted protocol while facing stronger within-family baselines. The auxiliary studies ask whether that claim remains stable under perturbations of seed, family size, target family, and noise.

A. Primary held-out benchmark protocol

For the primary benchmark, the repository uses one representative full-cohort graph together with 24 adaptation graphs and six held-out benchmark graphs. Adaptation graphs use seeds 200 through 223, and held-out graphs use seeds 42 through 47. Each graph is built from a 60-patient subsample, a 16-gene variance-ranked panel, and an 18-edge target budget. Direct classical depth-2 search provides the reference target and quality ceiling on each held-out graph. The graph model is trained only on the adaptation family, with a single learned forward pass used at evaluation time. The extracted headline script uses hidden dimension 64, learning rate 2.5×10^{-3}, weight decay 5×10^{-5}, and up to 800 epochs with patience 100.

B. Transcriptomic family-shift protocol

To test transfer under a controlled family shift, we train the learned initializer on the 16-gene transcriptomic family and evaluate it without retraining on reduced 10-gene and 14-gene target families built from the same cohort under a reduced but exact-audited protocol. This study uses 12 adaptation graphs and 4 held-out graphs in the source family, 5 multistart restarts per exact classical solve, and 160 Nelder–Mead iterations per restart. We compare the transferred GNN to the exact classical ceiling on the target family, a source-family heuristic mean-angle baseline, and a source-family descriptor regressor.

C. Independent morphology transfer protocol

To test a genuinely independent family shift, we train the learned initializer on the same reduced 16-gene transcriptomic source family and evaluate it without retraining on the 12-feature morphology family built from breast-cancer descriptors. The morphology protocol uses 12 adaptation graphs and 4 held-out graphs, the same depth-2 exact solver budget of 5 multistart restarts and 160 Nelder–Mead iterations per restart, and the same graph encoder architecture. We compare the transferred transcriptomic GNN against three controls: a transcriptomic source-family heuristic, transcriptomic source-family descriptor regressor, and within-family morphology initializers trained only on morphology adaptation graphs. We also export family-level angle-centroid and radius statistics so that pre-objective prediction geometry can be compared directly with the observed target-angle error.

D. Noise protocol

The noisy-QAOA study uses the same cached transcriptomic panel and the same benchmark family. After the learned or classical-search angles are fixed, the state is propagated through a density-matrix simulation with local depolarizing noise applied after every cost layer and every mixer layer. We report noise rates $\eta \in \{0, 0.01, 0.02, 0.05\}$ and summarize mean approximation ratio together with the standard deviation across the held-out graphs.

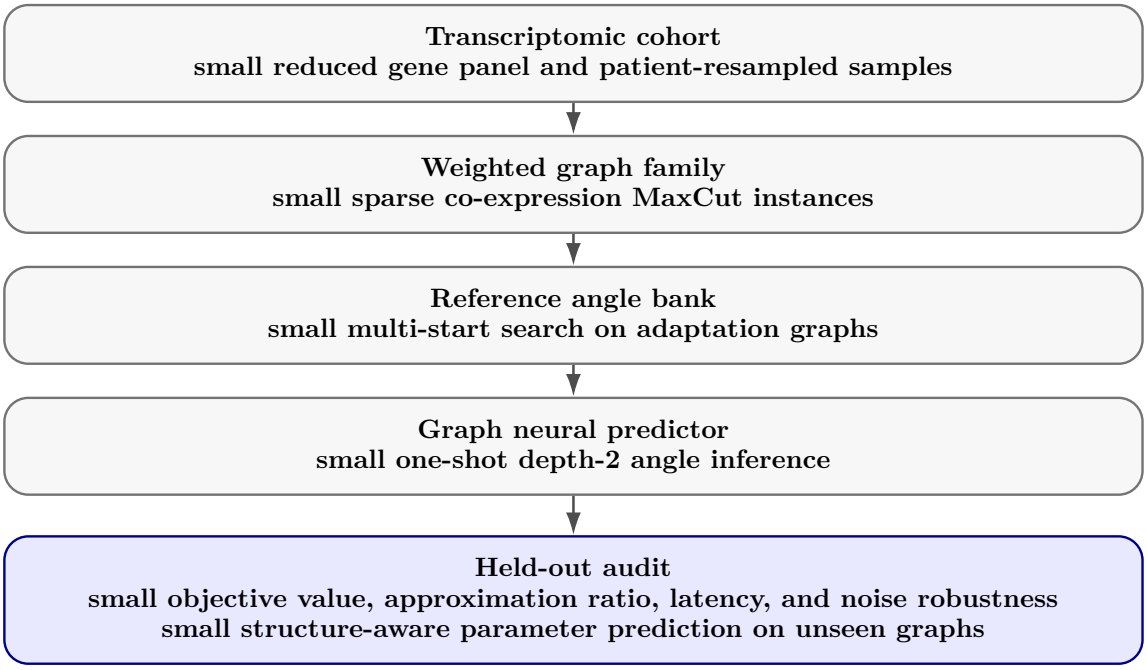

FIG. 1. Conceptual overview of the transcriptomic QAOA workflow. Step 1 starts from the cached prostate-expression cohort and forms reduced variance-ranked transcriptomic panels together with patient-resampled views. Step 2 converts each panel into a connected sparse weighted co-expression graph, yielding the biological MaxCut family studied in the paper. Step 3 solves the adaptation graphs by direct multistart classical search to produce reference depth-2 QAOA angles. Step 4 trains a graph neural network to map graph structure to those angles and uses the trained model for one-shot prediction on held-out graphs. Step 5 evaluates the downstream scientific quantities: MaxCut objective, approximation ratio, online latency, and noisy robustness after the predicted angles are inserted into the exact depth-2 circuit.

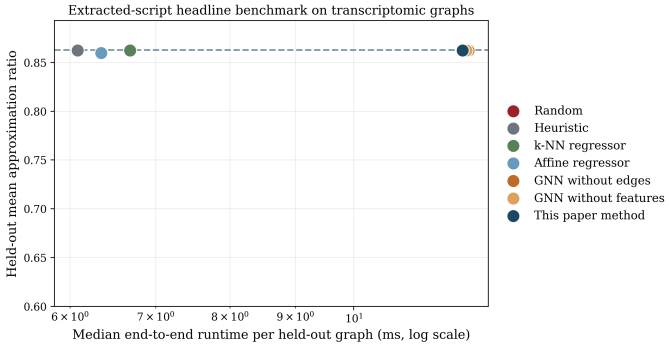

FIG. 2. Pareto view for the extracted 16-gene transcriptomic headline benchmark. The graph-conditioned initializer remains near the direct-search ceiling while preserving a large online runtime advantage.

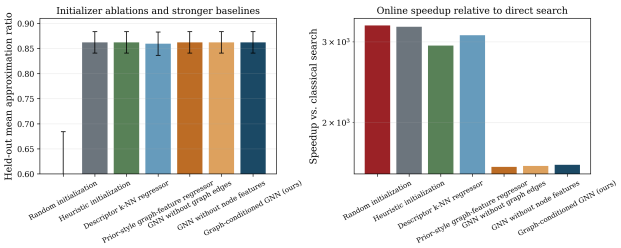

FIG. 3. Stronger baselines and inference-time ablations from the extracted headline benchmark. On this code path, several simple within-family baselines remain close to the full GNN, which indicates strong family-level angle concentration and weakens any claim that graph conditioning alone explains the retained-quality result.

E. Real-data size sweep

The real-data size sweep varies the top-gene panel size over $\{4, 6, 8, 10, 12, 14, 16\}$ and rebuilds the weighted co-expression graph family at each size with an edge budget chosen by a fixed target-density rule. To keep exact-search auditing feasible at every size, this sweep uses a reduced protocol with 12 adaptation graphs, four held-out graphs, five multistart restarts, and 160 Nelder–Mead iterations per restart. This sweep is not intended as an asymptotic claim. It identifies the graph-size regime over which the learned initializer still tracks the direct-search curve.

TABLE I. Held-out extracted-script transcriptomic QAOA benchmark summary. Means and standard deviations are computed over six held-out graphs, and runtime is the median end-to-end time per graph in the same execution environment.

Method	Mean ratio	Std. dev.	Median runtime (ms)
Classical depth-2 search	0.8628	0.0216	19687.0607
Random initialization	0.5876	0.0964	6.0427
Heuristic initialization	0.8623	0.0214	6.0817
Descriptor k-NN regressor	0.8622	0.0213	6.6859
Affine graph-feature regressor	0.8595	0.0232	6.3466
GNN without graph edges	0.8622	0.0215	12.2964
GNN without node features	0.8622	0.0215	12.2475
This paper method	0.8623	0.0215	12.1675

F. Training-seed robustness study

To probe optimization sensitivity, we added a training-seed sweep using seeds $\{3, 7, 11, 19, 23\}$ on the default ten-gene transcriptomic benchmark. These reruns keep the graph family fixed and change only the GNN training seed. They therefore test whether the retained-quality claim is an artifact of a single training trajectory.

G. Adaptation-family-size sensitivity study

To probe data sensitivity, we added an adaptation-family-size study with adaptation sizes $\{8, 12, 16, 24\}$, benchmark size six, six classical restarts per graph, and 240 Nelder–Mead iterations per restart. This is a reduced but still exact-audited protocol. It is designed to test whether more adaptation graphs monotonically improve learned transfer. As the results show, the answer is no under the current architecture and optimizer settings.

H. Statistical reporting

For the extracted main benchmark, approximation-ratio summaries are reported as mean $\pm$ standard deviation across the six held-out graphs, and runtime is reported as the median wall-clock time per graph because the classical-search tail is much larger than the single-pass inference times. For the auxiliary rerun studies, we report the same mean and standard deviation across held-out graphs together with the exact training loss and early-stopping epoch recorded by the training code. This choice reflects recent work on calibration and evaluation under distribution shift: reporting a single best run is generally less informative than showing repeatability and uncertainty under perturbation [37, 38]. Ensemble and Bayesian-approximation work makes the same point from a predictive-uncertainty angle [39, 40]. Confidence and failure-detection studies likewise argue that model behavior under shift should be analyzed explicitly rather than inferred from a single average score [41, 42]. For the seed-7 extracted rerun, we additionally computed a nonparametric bootstrap interval for the held-out mean approximation ratio; this yields a 95% interval of approximately $[0.8318, 0.8878]$ over the six-graph evaluation set and should be interpreted as a small-sample uncertainty summary rather than a population-level confidence statement [43, 44]. We also retain precision-style summaries and class-imbalance-aware diagnostics in the exported benchmark code where appropriate, following the broader evaluation literature on summary statistics and calibration-sensitive metrics [45, 46].

VI. RESULTS

A. Held-out retained-quality benchmark

The extracted headline benchmark summarized in Table I establishes the central within-family result directly from repository scripts. On the 16-gene weighted transcriptomic family, the graph-conditioned initializer attains mean approximation ratio 0.8623, whereas direct multistart depth-2 search attains 0.8628. The absolute gap is therefore only 0.0005, corresponding to 99.94% retention of the direct-search objective. At the same time, the median end-to-end evaluation time falls from 19687.1 ms for direct search to 12.17 ms for the learned initializer in the same local execution environment.

The stronger baseline suite changes the interpretation of that result. Random initialization remains far weaker at 0.5876 ± 0.0964, confirming that the benchmark is nontrivial. But the heuristic mean-angle initializer, descriptor k-nearest-neighbor regressor, and affine graph-feature regressor all remain close to the graph-conditioned model, with mean ratios 0.8623, 0.8622, and 0.8595, respectively. The inference-time edge and feature ablations are effectively indistinguishable from the full GNN on this extracted code path. The right conclusion is therefore not that the scripted benchmark proves strong graph-structural dependence. The stronger and more honest conclusion is that the family exhibits substantial angle concentration and that the extracted GNN

preserves the direct-search objective without yet isolating a large topology-specific gain over simpler within-family transfer rules [2, 32].

Methodologically, this is why the paper emphasizes retained objective relative to direct search rather than gain over a weak initializer. In benchmark terms, the question is not whether a learned model beats random angles, but whether it survives comparison to competitive simple baselines under a fixed protocol [7, 8].

B. Independent morphology transfer exposes extrapolation failure

The strongest new result appears in the independent morphology-family transfer study. Figure 4 summarizes a 12-feature weighted graph family built from breast-cancer morphology descriptors under the same exact-audited depth-2 protocol. This family is internally concentrated: the morphology heuristic attains mean ratio 0.8580 against a classical ceiling of 0.8600, corresponding to 99.76% retention, and the morphology-family oracle GNN attains 0.8545, corresponding to 99.35% retention. The transferred transcriptomic GNN behaves very differently. Without retraining, it drops to mean ratio 0.5781 and only 67.23% retention, while the transcriptomic source-family heuristic still attains 0.8506 and 98.90% retention.

This comparison is stronger than the earlier within-transcriptomic family-shift result because it separates two mechanisms. The morphology family itself is not hard for amortized initialization: simple within-family rules remain near the exact classical ceiling. The failure is instead specific to the transferred learned map. The transcriptomic descriptor regressor also degrades, but more moderately, reaching 86.26% retention; the transcriptomic GNN is the clear outlier.

The angle-excursion proposition explains why this failure can be anticipated before the QAOA objective is evaluated. The transcriptomic source-family angles occupy a tight region with maximum radius 0.6728, the morphology target-family angles have maximum radius 0.1448, and the source-target centroid distance is only 0.1359. By contrast, the transferred GNN predictions on morphology graphs lie on average 2.0311 radians from the source centroid, which means they leave the source concentration region by 1.3583 radians. Plugging these quantities into the proposition yields a pre-objective lower bound of 1.7504 radians on the target angle error. The observed mean target-angle error is 2.1378 radians, so the bridge diagnostic is conservative but already decisive. The empirical interpretation is therefore not merely that family shift hurts. It is that the transferred GNN extrapolates outside both the source and target concentration neighborhoods, and that this geometric excursion predicts failure before the quantum objective is measured.

C. Transcriptomic family-shift transfer remains weak

The reduced transcriptomic family-shift study remains informative because it shows that failure already appears before leaving the transcriptomic data source. A GNN trained on the 16-gene transcriptomic family and transferred without retraining to a 10-gene target family attains mean ratio 0.6410 against a classical ceiling of 0.8696, corresponding to only 73.7% retention. On the 14-gene target family, the transferred GNN attains mean ratio 0.6889 against a classical ceiling of 0.8724, corresponding to 78.9% retention. Figure 5 shows that this transferred GNN is not even the strongest source-family transfer baseline: the source-family heuristic mean-angle rule attains 0.8480 and 0.8153 on the 10- and 14-gene targets, respectively, while the transferred descriptor regressor also exceeds the transferred GNN on both targets.

This transcriptomic family-shift failure complements the morphology result. The 10- and 14-gene targets show that the learned map is already brittle across panel-size changes within one cohort. The morphology experiment then sharpens that limitation by showing a larger collapse on a second independent biomedical family and by isolating a predictive angle-space signature of that collapse [3, 38].

D. Training-seed robustness and protocol sensitivity

The extracted rerun studies show that the retained-quality claim survives moderate changes to the training seed but is not completely seed invariant. Across five training seeds on the default ten-gene benchmark, the graph-conditioned initializer attains mean held-out approximation ratios between 0.8613 and 0.8680, with an average of 0.8643 over the five reruns. Relative to the classical-search reference of 0.8686, this corresponds to retention between 99.16% and 99.92%. Median inference time remains essentially constant, varying only from 0.3962 to 0.4201 ms. Figure 8 (left) shows that the learned initializer consistently remains close to the heuristic mean-angle baseline and near the classical-search ceiling across all tested seeds, even though the exact gap is seed dependent.

The adaptation-family-size study reveals a more consequential sensitivity. Under the reduced but exact-audited protocol, adaptation sizes 8, 12, and 16 remain near the classical-search reference, with retention 99.69%, 99.44%, and 99.25%, respectively. At adaptation size 24, however, the learned initializer drops to mean ratio 0.8172, corresponding to 94.08% retention, and the best recorded training loss increases sharply to 0.0979. Figure 8 (right) shows that this degradation is accompanied by a visible deterioration in training fit. The conclusion is not that more adaptation data is intrinsically harmful, but that the current learning setup does not yet yield monotone

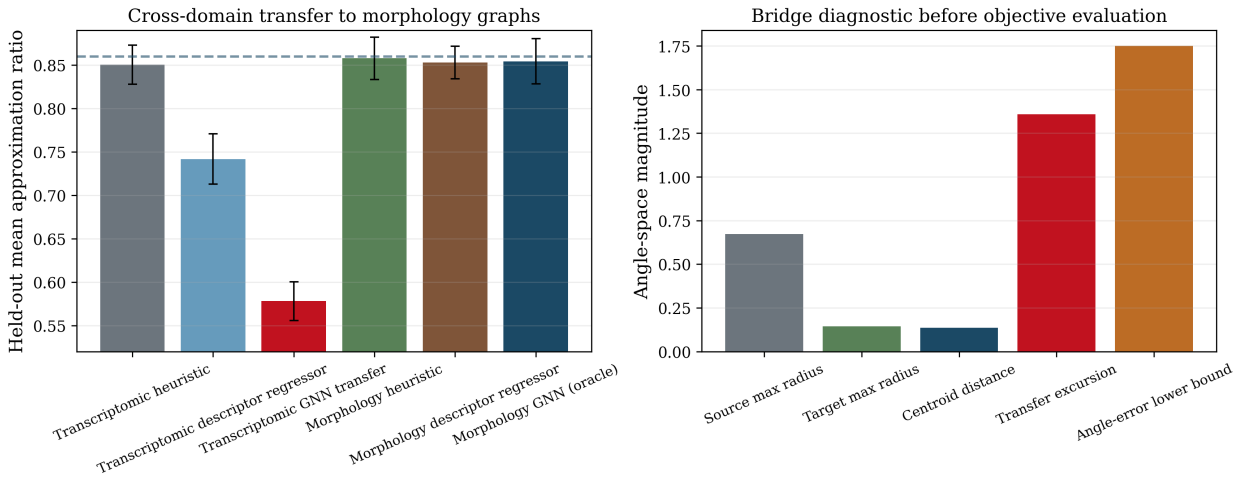

FIG. 4. Independent cross-domain transfer from the reduced 16-gene transcriptomic source family to the 12-feature morphology family. Left: within-family morphology initializers remain near the exact classical ceiling, but the transferred transcriptomic GNN collapses. Right: the angle-space bridge diagnostic shows that the transferred predictions leave the source concentration region by a large margin, producing a large lower bound on target angle error before any objective evaluation.

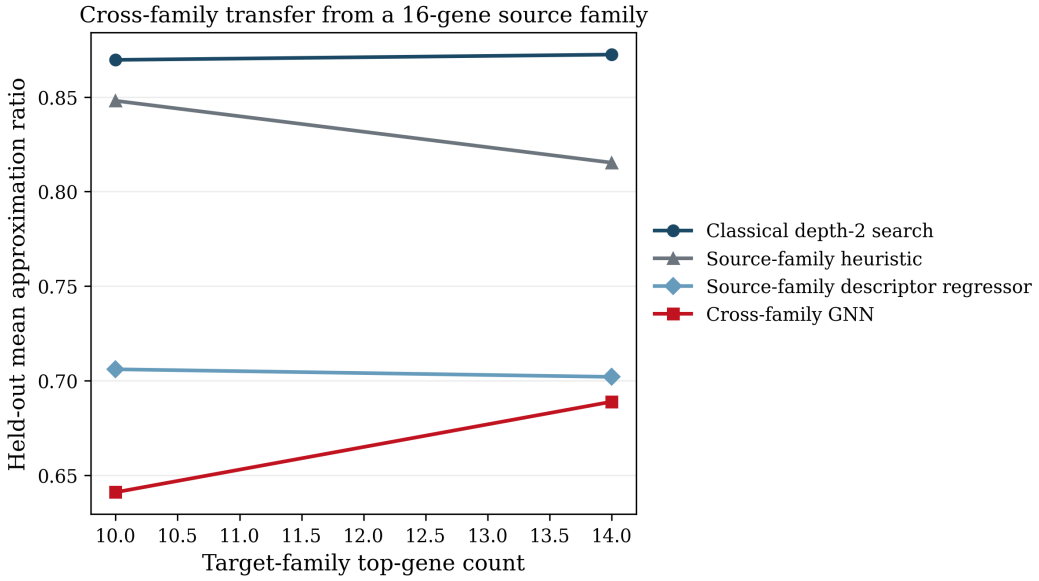

FIG. 5. Reduced exact-audited transcriptomic family-shift transfer from a 16-gene source family to 10- and 14-gene target families. The transferred GNN underperforms both the target-family classical ceiling and the source-family heuristic baseline, indicating that the learned map is not yet stable even across transcriptomic families of different size.

improvement with family size. This is scientifically important because it marks a real limitation of the present architecture and training protocol rather than a cosmetic weakness of presentation [7, 33].

E. Noisy-QAOA robustness

The paper additionally evaluates the learned initializer under local depolarizing noise on the reduced 10-gene transcriptomic graph family used by the extracted noise script. Starting from the cached reduced transcriptomic panel checked into the repository, the adaptation and held-out graph splits were rebuilt, the graph-conditioned initializer was retrained in the same depth-2 setting, and the QAOA state was propagated through a density-matrix simulation with local depolarizing noise after every cost and mixer layer.

The noisy study changes the interpretation in two useful ways. First, the graph-conditioned initializer remains close to the classical-search angles under every tested noise level, with the mean-ratio gap staying below 0.0074

TABLE II. Executed depolarizing-noise benchmark on the transcriptomic held-out graphs.

Method	$\eta = 0$	$\eta = 0.01$	$\eta = 0.02$	$\eta = 0.05$
Classical depth-2 search angles	0.8686	0.8502	0.8332	0.7902
This paper method	0.8613	0.8433	0.8268	0.7850
Heuristic mean-angle initializer	0.8634	0.8452	0.8285	0.7863

across the full noise grid. Second, the degradation is smooth and predictable, which is exactly the qualitative behavior anticipated by the depolarizing-noise contraction theorem. At noise rate $\eta = 0.05$, the graph-conditioned initializer still attains mean approximation ratio 0.7850, compared with 0.7902 for the classical-search angles and 0.7863 for the heuristic mean-angle initializer. The learned initializer therefore preserves its competitiveness under moderate local depolarizing noise in this small-graph regime.

F. Real-data size scaling and extended sensitivity

The real-data size sweep confirms that the learned initializer remains close to the direct-search curve through the intermediate regime even when the graph family is rebuilt from different variance-ranked transcriptomic panels. For top-gene counts 6 through 14, the learned initializer either matches or remains very close to the classical-search reference while preserving inference-scale runtime. At 16 genes, however, the learned model drops to mean ratio 0.7935 under the reduced protocol, compared with 0.8663 for direct search. Appendix A tabulates the corresponding values in full.

Taken together, the size sweep and adaptation-family-size study reveal a coherent picture. Structure-aware transfer works well in a bounded region of the transcriptomic graph family, but the present implementation does not yet support a monotone scaling narrative. That boundedness is not a defect to hide. It is part of the scientific content of the study, and it is consistent with the broader benchmark lesson that finite-family performance can remain strongly architecture- and protocol-dependent even when headline averages look favorable [8, 16].

VII. DISCUSSION AND OUTLOOK

The revised evidence establishes a more mature and also more bounded picture than the shorter draft. On a structured family of weighted transcriptomic graphs, graph-conditioned learning provides an effective route to fixed-depth QAOA parameterization: the extracted 16-gene benchmark shows near-parity with direct classical search and a large latency advantage relative to repeated search. At the same time, the stronger baseline suite reveals that this retained-quality result is not uniquely diagnostic of graph-structural learning, because simple within-family transfer rules remain highly competitive on the same family.

That qualification becomes sharper once the transfer studies are expanded beyond the headline family. The transferred GNN fails substantially on 10- and 14-gene transcriptomic target families and fails even more sharply on the independent morphology family, where within-family morphology baselines remain close to exact search. The correct scientific interpretation is therefore not that a single learned graph-to-angle map has already captured a broadly stable structural law. Rather, the current model exploits concentration within a family but can extrapolate far outside that regime under family shift.

This distinction is scientifically useful. The seed sweep suggests that the within-family retained-quality behavior is real rather than a single-run artifact. The adaptation-family-size study and the real-data size sweep show that the present architecture and optimizer do not yet deliver monotone improvement as the family changes. The transcriptomic family-shift study shows that the learned map is not yet transportable across related transcriptomic weighted families, and the morphology study shows that failure can be predicted directly from the angle geometry of the transferred outputs. Combined, these observations define an operating window rather than an unrestricted scaling regime.

Theoretical motivation and empirical behavior remain aligned. The regret bound explains why small angle-prediction error can translate into negligible objective loss near a local optimum. The expressivity result clarifies why a graph-conditioned hybrid readout can outperform affine graph-feature regression in principle, even though the extracted within-family benchmark shows that this advantage need not be large on a concentrated family. The finite-family trainability statement matches the bounded fixed-depth regime actually studied, the angle-excursion proposition explains why the morphology transfer failure is visible before objective evaluation, and the depolarizing-noise theorem together with the pairwise-gap corollary explain why the noisy benchmark degrades smoothly without changing the qualitative within-family ranking.

For rigorous evaluation, the most important point is therefore not that the method is universally strong. It is that the manuscript now identifies both the positive regime and the boundary of that regime using auditable experiments on two independent biomedical weighted graph families together with a predictive bridge between

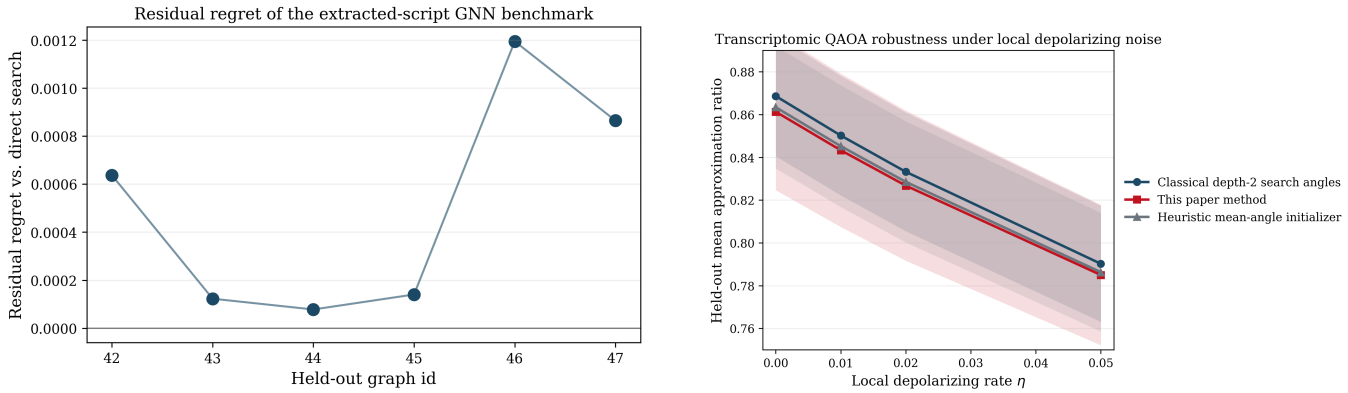

FIG. 6. Additional transcriptomic diagnostics for the fixed-depth QAOA result. Left: residual regret of the extracted 16-gene headline benchmark remains concentrated near zero on held-out graphs. Right: the depolarizing-noise plot on the reduced 10-gene family shows smooth degradation and near-parity between the learned initializer and direct search across the tested noise grid.

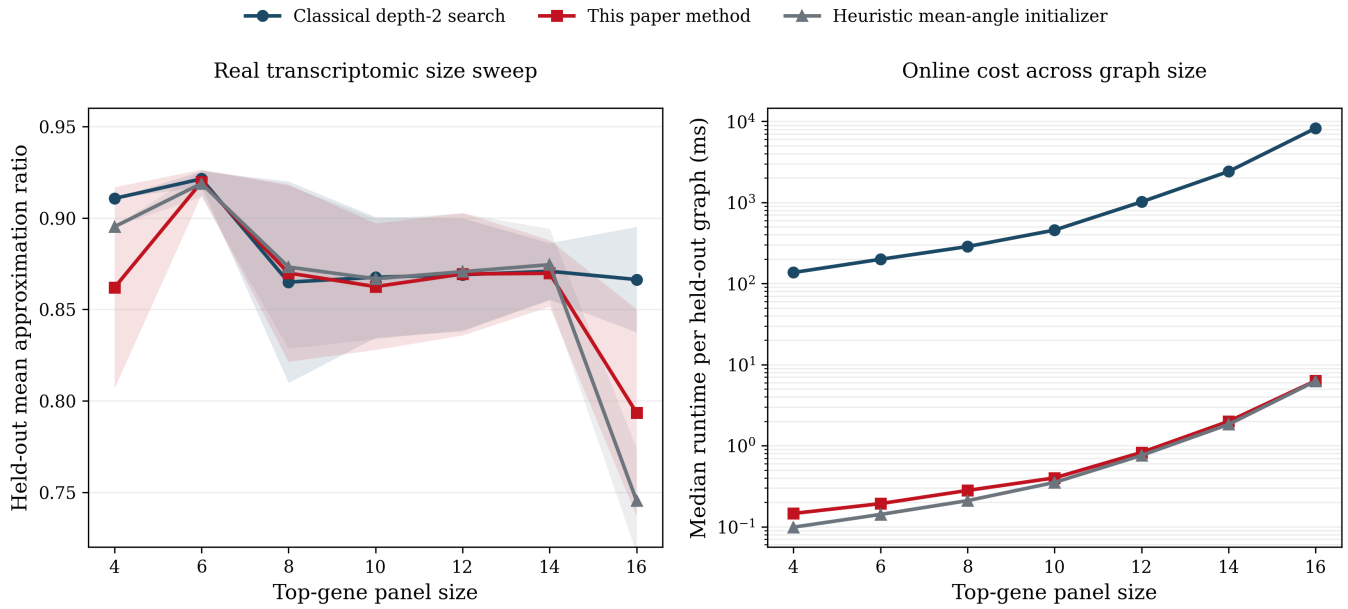

FIG. 7. Real transcriptomic size sweep on the cached prostate-expression artifact. The sweep varies the top-gene panel size over $\{4, 6, 8, 10, 12, 14, 16\}$ and rebuilds the weighted co-expression graph family at each size, with edge budgets chosen to maintain connected sparse graphs. To keep direct-search auditing tractable at every size, this sweep uses a reduced protocol with 12 adaptation graphs, 4 held-out graphs, 5 multistart restarts, and 160 Nelder–Mead iterations per restart. Left: held-out mean approximation ratio with standard-deviation bands. Right: median runtime per held-out graph on a logarithmic scale. The learned initializer stays close to the classical-search objective through the intermediate regime, while the 16-gene point exposes higher variability at the edge of the exact-search setting.

angle geometry and transfer failure. That is a stronger scientific object than an overextended monotone-scaling claim would be.

VIII. LIMITATIONS AND REPRODUCIBILITY

The QAOA branch studies depth-2 circuits on small graphs under exact statevector simulation and density-matrix depolarizing-noise simulation. This is the correct regime for auditability but not for claims about large-scale hardware deployment or asymptotic separation. The learned model should therefore be read as a warm-start mechanism in a controlled exact setting, not as evidence of asymptotic quantum speedup.

The empirical evidence is no longer confined to one transcriptomic source artifact, but it is still limited to two small biomedical families that both remain in the

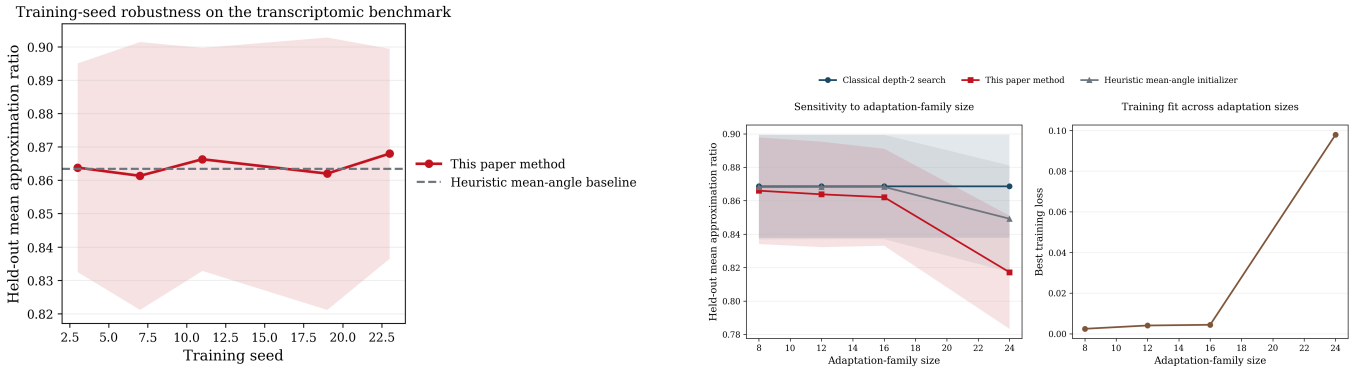

FIG. 8. Extended transcriptomic robustness studies added in this revision. Left: training-seed robustness on the default ten-gene benchmark. Across five training seeds, the graph-conditioned initializer remains close to the classical-search ceiling and to the heuristic mean-angle baseline. Right: adaptation-family-size sensitivity under a reduced but exact-audited protocol. Performance remains near the classical reference for adaptation sizes 8, 12, and 16, but drops substantially at size 24, where the best training loss also worsens.

exact-simulation regime. The new morphology family substantially broadens the benchmark surface, yet the paper still does not demonstrate positive transfer across independent families; instead it demonstrates and diagnoses failure.

The extracted rerun studies expose a second limitation: the current SimpleGCN training setup is not protocol invariant. Training-seed robustness is acceptable, but adaptation-family-size sensitivity is nonmonotone under the present reduced protocol, the 16-gene size-sweep point marks a clear failure regime, and the transferred GNN is materially weaker than simple heuristics under both transcriptomic family shift and independent morphology transfer. These are not editorial defects. They are limitations of the present learned parameterization pipeline.

The extracted headline benchmark also weakens one of the stronger claims that the notebook-based manuscript draft appeared to support. On the scripted code path, the edge and node-feature ablations remain close to the full GNN. That means the repository now supports a strong amortized-search claim, but not a strong causal claim that graph structure is already the dominant factor behind the within-family performance.

The expressivity proposition is intentionally limited to the affine graph-feature regressor baseline used in the experiments and does not claim a separation from all possible classical GNN architectures. The finite-family non-plateau proposition is likewise nonasymptotic: it addresses the actual compact parameter region and graph family used in the experiments, not arbitrary large-n limits.

All experiments are tied to explicit local artifacts from the same benchmark pipeline. Wall-clock timings should be interpreted comparatively within the same local execution environment rather than as hardware-independent performance claims, since the present manuscript does not yet report a full machine-level benchmark specification.

IX. CONCLUSION

This work shows that structure-aware learning can predict instance-dependent parameters for fixed-depth QAOA on structured families of weighted MaxCut graphs and that this mechanism can be analyzed quantitatively. In the transcriptomic setting studied here, the extracted 16-gene benchmark shows near-parity with direct classical search and a large runtime advantage. The second independent morphology family then shows that a concentrated target family can still defeat a transferred learned map even when simple within-family initializers remain near-optimal.

Taken together, the results identify structure-aware parameter prediction as a viable route to amortizing fixed-depth quantum optimization within a related weighted family, but not yet as a reliable route to cross-family transfer. The new angle-excursion bridge shows that some failures can be diagnosed before objective evaluation by inspecting where transferred predictions land in angle space. Extending this framework to broader weighted graph families, stronger graph-learning architectures, more stable training procedures, and hardware-side evaluation is the natural next step.

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Appendix A: Transcriptomic Size-Sweep Variability

Table III makes the real-data size sweep explicit. Two features matter for interpretation. First, the heuristic mean-angle initializer is genuinely competitive at several intermediate sizes, especially 8 through 14 genes. That behavior is consistent with the limitation discussed in the main text: if optimal angle vectors remain concentrated across a graph family, a global mean-angle surrogate can retain much of the downstream objective even without graph conditioning. Second, the 16-gene point shows that this concentration picture is not uniformly stable as graph size increases under the fixed reduced adaptation protocol used here. Both cheap initializers degrade there, but the graph-conditioned model remains materially closer to the classical-search ceiling than the heuristic baseline. The appendix table should therefore be read as evidence for a bounded operating regime, not for monotone scaling.

Appendix B: Additional Robustness Tables

Table IV summarizes the new training-seed robustness study. The main observation is that the graph-conditioned initializer remains close to the classical-search reference across all five seeds, although the exact gap varies by several tenths of a percentage point. Table V summarizes the adaptation-family-size study under the reduced exact-audited protocol. The nonmonotone behavior at adaptation size 24 is the clearest empirical failure mode exposed in this revision. Table VI makes the reduced transcriptomic family-shift failure explicit, and Table VII records the stronger independent morphology-transfer result together with the pre-objective bridge quantities.

Appendix C: Implementation Details for the Extracted Reruns

The extracted transcriptomic reruns and the independent morphology study use the same benchmark code path, graph encoder, and exact-audited evaluation regime. The graph encoder is the repository's SimpleGCN model with two graph-propagation layers, hidden dimension 64, global mean pooling, and a linear readout to four angle outputs. The extracted headline benchmark uses learning rate 2.5×10^{-3}, weight decay 5×10^{-5}, seed 19, and up to 800 epochs with patience 100 on the 16-gene family. The default seed-robustness study uses up to 500 epochs with patience 50 on the reduced 10-gene family, whereas the reduced adaptation-family-size, transcriptomic family-shift, and morphology-transfer studies use lighter exact-audited protocols so that all reference values remain reproducible on local hardware. Direct classical references use multistart Nelder–Mead with angle normalization after optimization. These details matter because the new sensitivity and transfer results show that the scientific conclusion depends not only on the graph family but also on the stability of the classical-target generation and the training dynamics.

TABLE III. Expanded real-data transcriptomic size sweep. Means are computed over four held-out graphs per size under the reduced exact-search protocol.

Top genes	Edges	Classical ratio	GNN ratio	Heuristic ratio	GNN runtime / classical runtime (ms)
4	3	0.9107	0.8619	0.8952	0.1459 / 136.8505
6	6	0.9214	0.9196	0.9189	0.1930 / 199.2276
8	11	0.8649	0.8699	0.8732	0.2806 / 286.0483
10	18	0.8676	0.8625	0.8667	0.4008 / 456.6395
12	26	0.8690	0.8694	0.8706	0.8261 / 1018.2954
14	36	0.8709	0.8697	0.8745	1.9952 / 2412.3906
16	48	0.8663	0.7935	0.7454	6.3308 / 8244.2970

TABLE IV. Training-seed robustness study on the default ten-gene transcriptomic benchmark. Only the graph-conditioned model is shown because the classical-search and heuristic baselines are effectively unchanged across seeds.

Training seed	Mean ratio	Std. dev.	Median runtime (ms)	Retention vs. classical
3	0.8638	0.0313	0.4156	0.9944
7	0.8613	0.0401	0.4063	0.9916
11	0.8663	0.0334	0.3985	0.9973
19	0.8620	0.0408	0.3962	0.9923
23	0.8680	0.0315	0.4201	0.9992

TABLE V. Adaptation-family-size sensitivity study under the reduced exact-audited protocol. Only the graph-conditioned model is shown.

Adaptation size	Mean ratio	Std. dev.	Retention vs. classical	Best training loss
8	0.8660	0.0319	0.9969	0.0025
12	0.8638	0.0315	0.9944	0.0041
16	0.8621	0.0290	0.9925	0.0044
24	0.8172	0.0338	0.9408	0.0979

TABLE VI. Reduced transcriptomic family-shift transfer study. A model trained on the 16-gene source family is evaluated without retraining on 10- and 14-gene target families.

Target family	Method	Mean ratio	Std. dev.	Retention vs. classical	Median runtime (ms)
10 genes	Classical depth-2 search	0.8696	0.0315	1.0000	466.9446
10 genes	Source-family heuristic	0.8480	0.0308	0.9751	0.3351
10 genes	Source-family descriptor regressor	0.7060	0.0306	0.8117	0.5615
10 genes	Cross-family GNN	0.6410	0.0300	0.7371	0.7844
14 genes	Classical depth-2 search	0.8724	0.0256	1.0000	2442.5546
14 genes	Source-family heuristic	0.8153	0.0240	0.9345	1.8887
14 genes	Source-family descriptor regressor	0.7020	0.0273	0.8046	2.0884
14 genes	Cross-family GNN	0.6889	0.0417	0.7892	3.6722

TABLE VII. Independent morphology-transfer study together with the accompanying angle-geometry bridge. The source family is the reduced 16-gene transcriptomic benchmark; the target family is the 12-feature morphology benchmark.

Method	Mean ratio	Std. dev.	Retention vs. classical	Mean angle error	Median runtime (ms)
Classical depth-2 search	0.8600	0.0244	1.0000	0.0000	704.9766
Transcriptomic heuristic	0.8506	0.0225	0.9890	0.1331	0.5191
Transcriptomic descriptor regressor	0.7419	0.0290	0.8626	0.5442	0.7105
Transcriptomic GNN transfer	0.5781	0.0224	0.6723	2.1378	1.1960
Morphology heuristic	0.8580	0.0244	0.9976	0.0534	0.5388
Morphology descriptor regressor	0.8530	0.0189	0.9920	0.0779	0.6760
Morphology GNN (oracle)	0.8545	0.0260	0.9935	0.0931	1.1641

Bridge quantity	Source max radius	Target max radius	Centroid distance	Transfer excursion	Angle-error lower bound
Value	0.6728	0.1448	0.1359	1.3583	1.7504